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CHAPTER 1: INTRODUCTION

Physical fitness and sports participation have become essential components of a healthy and fulfilling life. With the rise of sedentary lifestyles, the prevalence of non-communicable diseases such as obesity, hypertension, and diabetes has increased dramatically. This has brought fitness and wellness to the forefront of public health concerns. Governments, healthcare providers, and educators have increasingly recognized the role of regular physical activity in disease prevention, mental health improvement, and overall quality of life. In addition to physical health, regular exercise has been linked to improved cognitive function, stress management, and better sleep patterns.

This minor research project is grounded in a comprehensive survey conducted among 100 individuals from various demographic backgrounds. The survey covered questions related to age, gender, religion, income, education, physical activity habits, fitness goals, injury experiences, and recovery practices. The core objective of this study is to investigate how different demographic variables influence fitness Behaviour and lifestyle choices. Do younger people exercise more frequently? Does income level affect the choice of physical activity? Are women more inclined toward yoga and wellness-based practices while men prefer strength training? These are the types of questions this project aims to answer.

Statistical tools such as the Chi-square test are used in this study to determine the relationship between demographic variables and fitness behaviours. For example, the test can reveal whether there is a significant association between age group and frequency of exercise. Similarly, graphical tools like pie charts, bar graphs, and histograms have been used to present survey findings in a visually engaging manner. These tools help translate numerical data into meaningful insights, allowing the reader to easily interpret trends and patterns. The visualizations serve not only to support statistical conclusions but also to offer a broader picture of how fitness habits vary within the population.

The impact of the COVID-19 pandemic on fitness behaviour is another focal point of this research. Lockdowns and gym closures disrupted routine physical activities for many, while simultaneously inspiring new approaches like home workouts and virtual fitness classes. This shift represents a significant cultural change in how fitness is perceived and practiced. The data collected through this project highlights these changes, giving us a glimpse into how the population adapted to maintain their health in challenging times. It also provides a sense of how resilient and health-conscious society has become post- pandemic.

CHAPTER 2: HISTORICAL REVIEW

The History of physical fitness and sports participation is rich and diverse, rooted deeply in human civilization. From ancient times, physical strength and agility were crucial for survival. Hunting, gathering, and defending territory required physical prowess, and over time, these skills evolved into structured activities and early forms of sport. Ancient societies such as Egypt, Greece, China, and India each had their own interpretations of physical activity, often blending them with religious or philosophical elements. Fitness, in those early ages, was not just a means to health but a way of life that carried deep cultural significance.

In ancient China, fitness was closely linked with medicine and philosophy. Practices such as Tai Chi and Qigong focused on internal energy, balance, and flexibility. The Chinese believed that physical activity helped regulate bodily functions and maintain harmony within the body. Martial arts, which later became formalized into distinct systems like Kung Fu, also played a major role in physical training and were used both for self- defence and spiritual development.

As civilizations progressed, the Middle Ages saw a decline in structured physical training, particularly in Europe, where intellectual and religious pursuits were prioritized over physical excellence. However, in other parts of the world, such as Japan, traditions like sumo wrestling and samurai martial arts flourished and were maintained through strict cultural rituals. The Renaissance period revived interest in human anatomy and physical education, particularly in Europe, where artists and scholars began to appreciate the human form and advocated for physical exercise.

The Industrial Revolution brought sweeping changes to society. Urbanization and mechanization reduced the physical demands of daily labour, leading to an increasingly sedentary lifestyle. However, the need for organized fitness programs became more apparent, especially as public health became a concern. During the 19th century, European countries like Germany and Sweden introduced structured physical education systems in schools, emphasizing gymnastics, military drills, and moral development.

In the 20th century, global awareness about fitness increased dramatically. The world wars demonstrated the importance of physical preparedness, leading to national fitness initiatives in several countries. The United States, for example, introduced the President's Council on Physical Fitness to promote nationwide participation in exercise.

CHAPTER 3: OBJECTIVES

1. To Examine the frequency and types of sports and fitness activities among different age groups.
2. To Analyse the relationship between fitness habits and demographic factors such as age, gender, and occupation.
3. To Identify trends and patterns in physical activity levels based on survey responses.
4. To Evaluate the impact of regular physical activity on self-reported health status.

CHAPTER 4: DATA COLLECTION

The Primary Instrument used for data collection in this project was a well-structured Survey **Questionnaire**. It was designed to collect information on various demographic and behavioural parameters such as age, gender, education level, occupation, physical activity habits, and preferred forms of fitness, duration of exercise, motivation for physical activity, dietary patterns, and changes in fitness behaviour post-COVID-19. The Questionnaire consisted of both closed-ended and multiple-choice questions, allowing for easy tabulation and Quantitative Analysis.

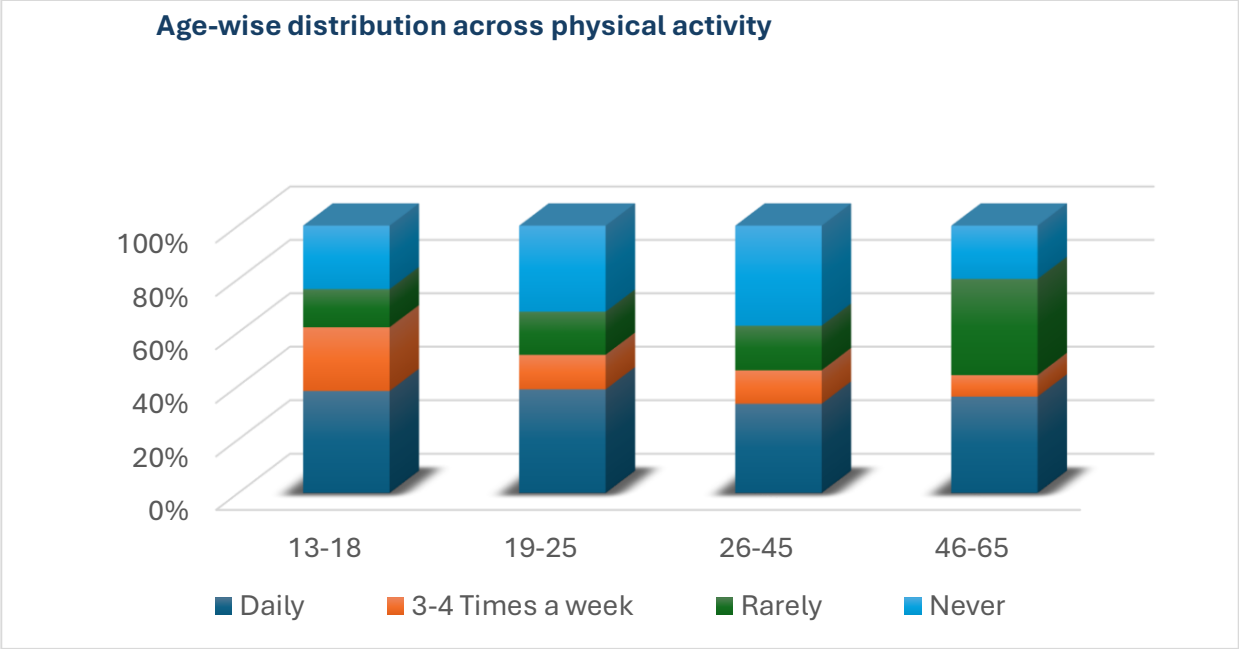
A total of **100 Respondents** were selected by **Convenience Sampling Method** for this survey-based study. The participants belonged to diverse age groups ranging from 18 to 51+ years. The data was collected through physical distribution of printed questionnaires, as well as online forms for remote participants. Care was taken to include a balanced sample in terms of gender and age to ensure representativeness and minimize sampling bias. The sampling method used was convenient sampling due to time constraints and accessibility.

The Data Collected from the participants was carefully reviewed and manually entered into spreadsheet software **(Microsoft Excel).** Responses were coded numerically to facilitate statistical analysis. For example, responses to the question “Do you exercise regularly?” were coded as 1 for 'Yes' and 0 for 'No'. Similarly, other categorical responses were also assigned numerical codes for ease of computation.

CHAPTER 5: DATA ANALYSIS

5.1 Age-wise distribution across physical activity: Engagement of people in Physical Activity according to their Age Group

Age Group	Daily	3-4 Times a week	Rarely	Never
13-18	8	5	3	5
19-25	12	4	5	10
26-45	8	3	4	9
46-65	9	2	9	5
Total	37	14	20	29

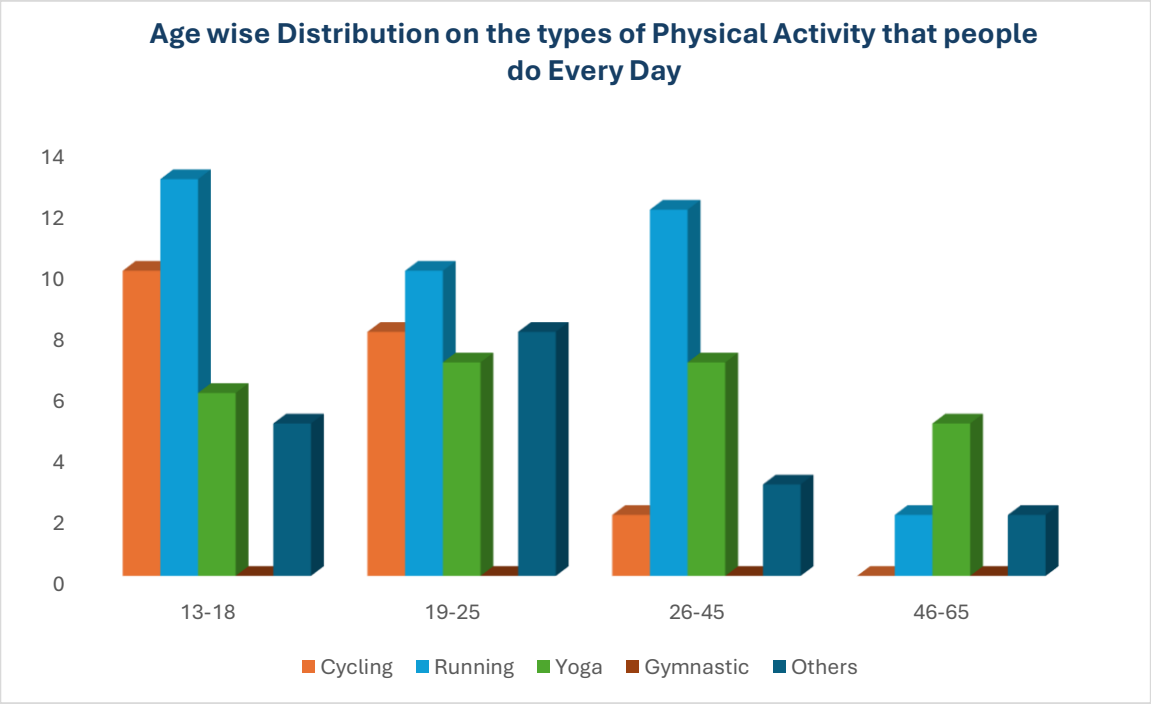


INTERPRETATIONS: -

- ✓ The Most active age group (Daily) - The 19-25 age group has highest no. of daily participants (12).
- ✓ Least active age group (Never) -The 19-25 group has also the highest no. of people who never exercise (10), showing polarization.
- ✓ Moderate engagement (3-4 times/week) -The 13-18 group is relatively balanced.
- ✓ Older adults (46-65) - Surprisingly, 9 out of 16 participants in 46-65 age group ex. daily, making them high.

5.2 Age wise Distribution on the types of Physical Activity that people do Every Day

Age Group	Cycling	Running	Yoga	Gymnastic	Others
13-18	10	13	6	0	5
19-25	8	10	7	0	8
26-45	2	12	7	0	3
46-65	0	2	5	0	2
Total	20	37	25	0	18

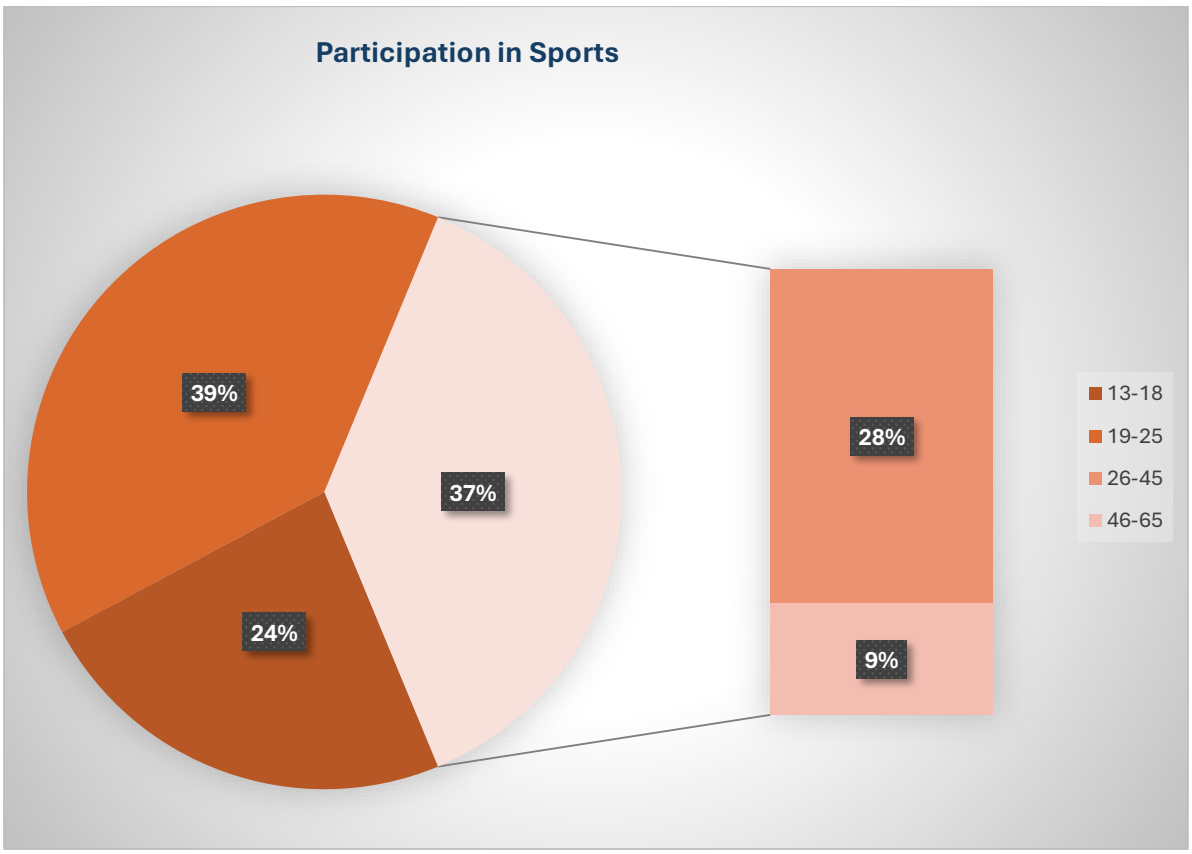


INTERPRETATIONS: -

- ✓ Most popular activity overall is Running as it is most common in all age groups.
- ✓ Least popular activity is Gymnastics.
- ✓ Yoga is consistently practiced across all age groups.
- ✓ Cycling is mostly done by younger groups (13-25)

5.3 Participation in Sports According to their Age Group

Age group	Yes	No
13-18	15	3
19-25	25	5
26-45	18	10
46-65	6	27
Total	64	45

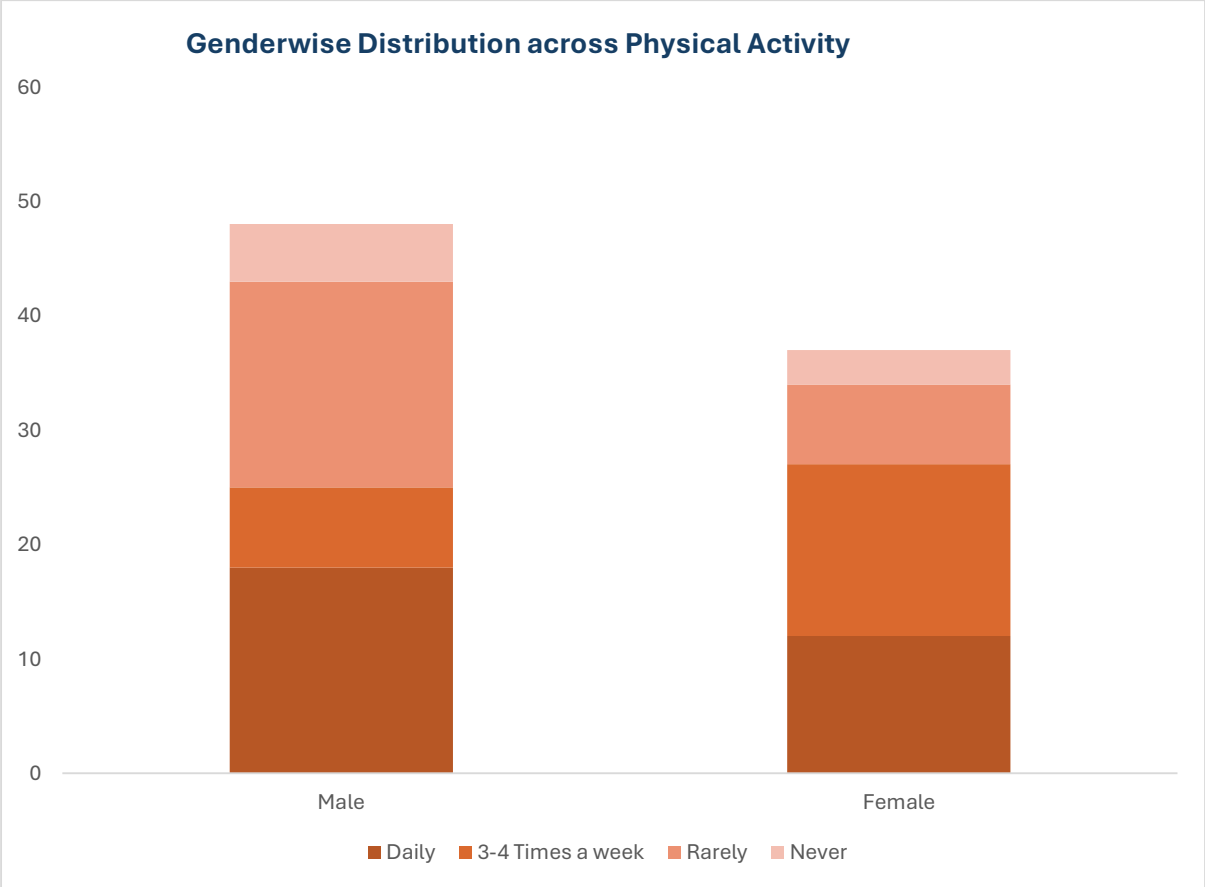


INTERPRETATIONS: -

- ✓ There is an active participation in Sports among all age groups.

5.4 Gender Wise Distribution across Physical Activity

Gender	Daily	3-4 Times a week	Rarely	Never
Male	18	7	18	5
Female	12	15	7	3
Total	30	22	25	8

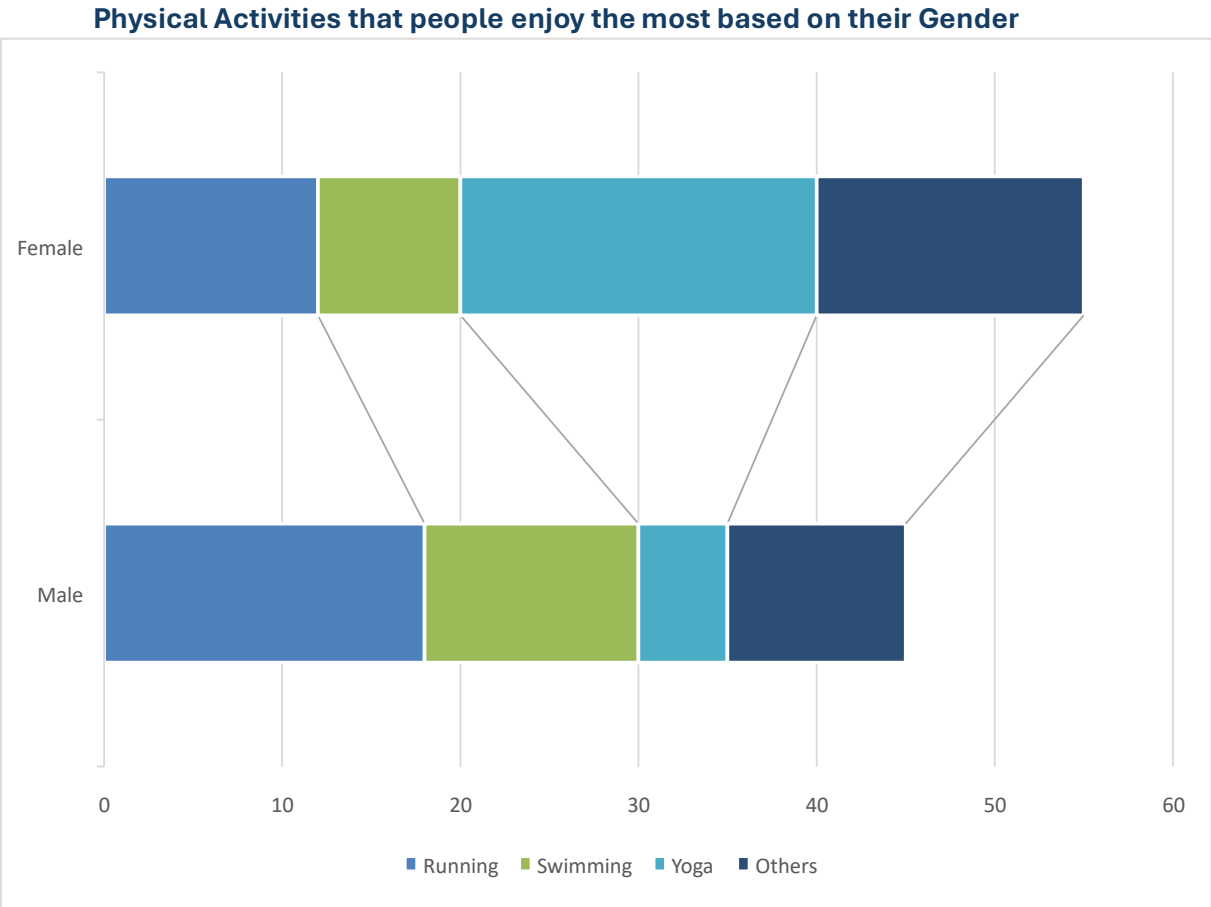


INTERPRETATIONS: -

- ✓ A majority of both Male & Female report daily exercise activity.
- ✓ Only females report never engaging in physical activity.
- ✓ Male appear slightly more active than female.
- ✓ Females show slightly more dispersion, with some reporting “Never”.

5.5 Types of Physical Activity that people enjoy the most based on their Gender

Total	Running	Swimming	Yoga	Others
Male	18	12	5	10
Female	12	8	20	15
Total	30	20	25	25

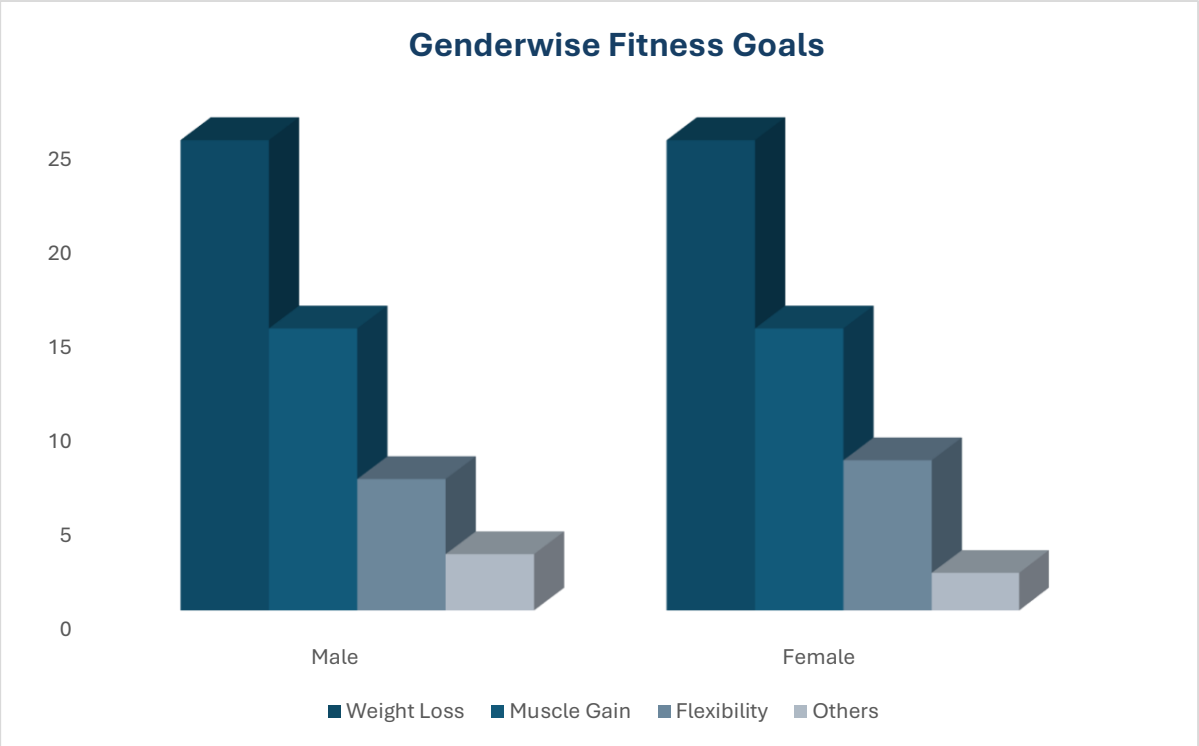


INTERPRETATIONS: -

- ✓ Running is slightly more preferred by males than females.
- ✓ Swimming participation is relatively balanced between genders.
- ✓ Yoga shows a higher preference among females compared to males.
- ✓ The category "Others" (e.g., gym, dance, etc.) also sees slightly higher participation from females.
- ✓ The data suggests a gender-based distribution in the types of physical activities people enjoy.

5.6 Fitness Goals of the People According to Gender

Gender	Weight Loss	Muscle Gain	Flexibility	Others
Male	25	15	7	3
Female	25	15	8	2
Total	50	30	15	5



INTERPRETATIONS: -

✓ Weight loss is equally prioritized by both genders.

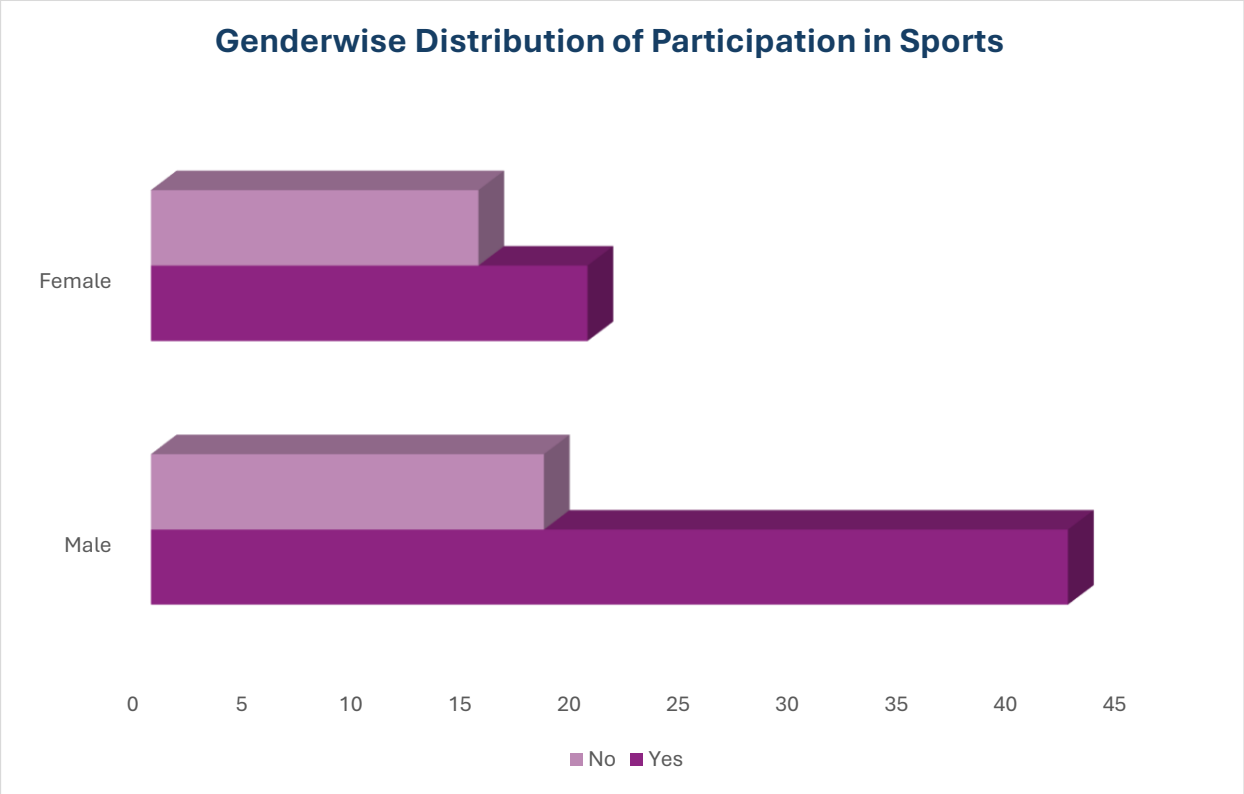
✓ Muscle gain shows equal interest observed by both genders.

✓ Improved flexibility sees slightly higher interest among females.

✓ The "Others" fitness goals show minor interest overall, with a slight preference among males.

5.7 Gender wise Distribution of participation in sports or exercise among the People

Gender	Yes	No
Male	42	18
Female	20	15
Total	62	33

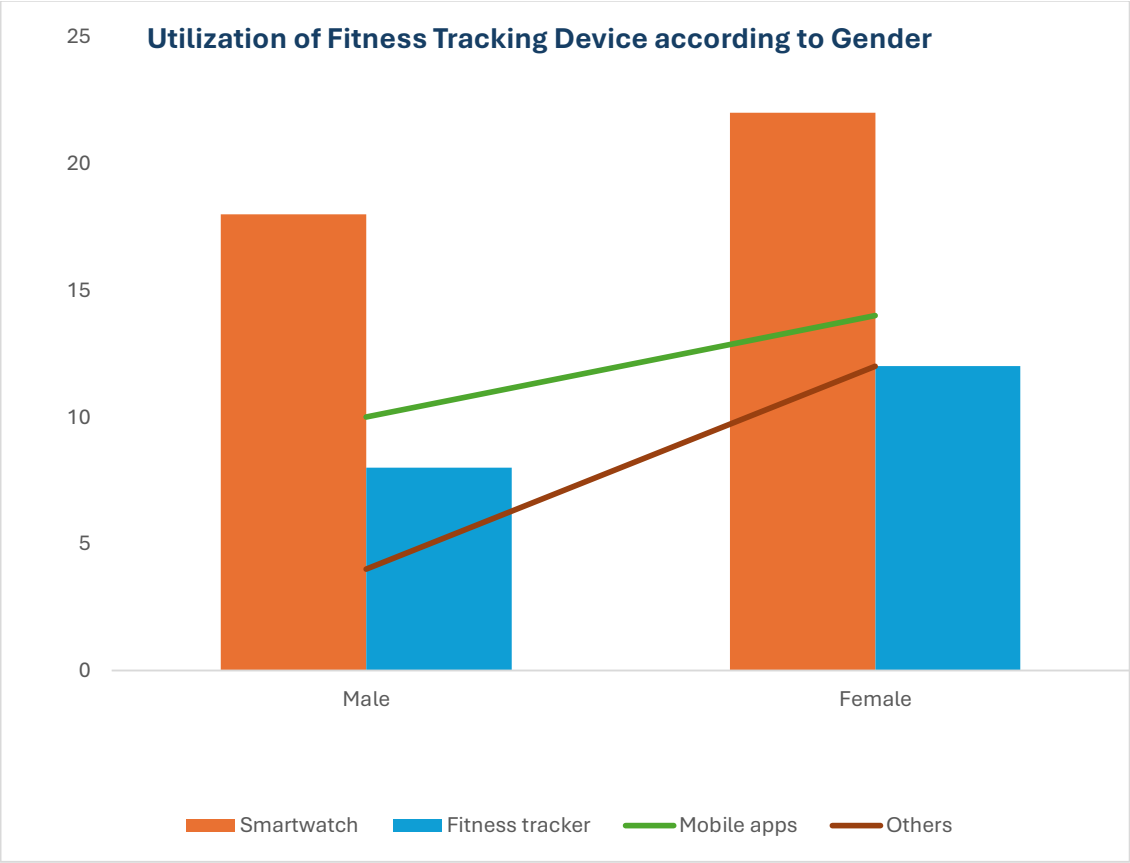


INTERPRETATIONS: -

- ✓ A higher proportion of males (70%) engage in sports compared to females (57%).
- ✓ The disparity may be influenced by various factors, including cultural norms, access to facilities and societal expectations.

5.8 Gender Distribution of Utilization of any Fitness Tracking Device or apps among People

Gender	Smartwatch	Fitness tracker	Mobile apps	Others
Male	18	8	10	4
Female	22	12	14	12
Total	40	20	24	16

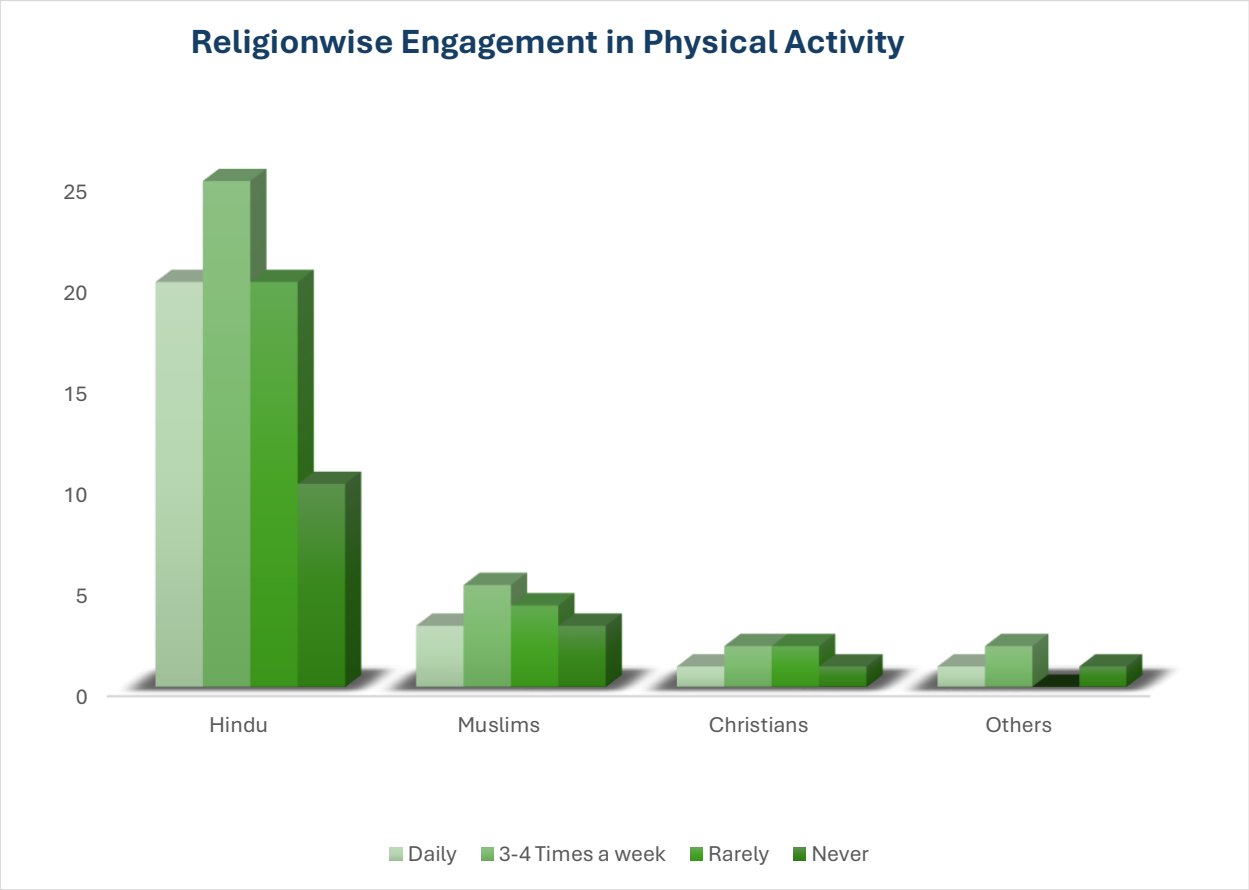


INTERPRETATIONS: -

- ✓ Higher Female Usage: Women tend to utilize fit tracking device and app more than males.
- ✓ Smart watch Preference: Both genders show a preference for smart watches, with women leading in usage.

5.9 Engagement in Physical Activity based on the Religion Distribution

Religion	Daily	3-4 Times a week	Rarely	Never
Hindu	20	25	20	10
Muslims	3	5	4	3
Christians	1	2	2	1
Others	1	2	0	1
Total	25	34	26	15

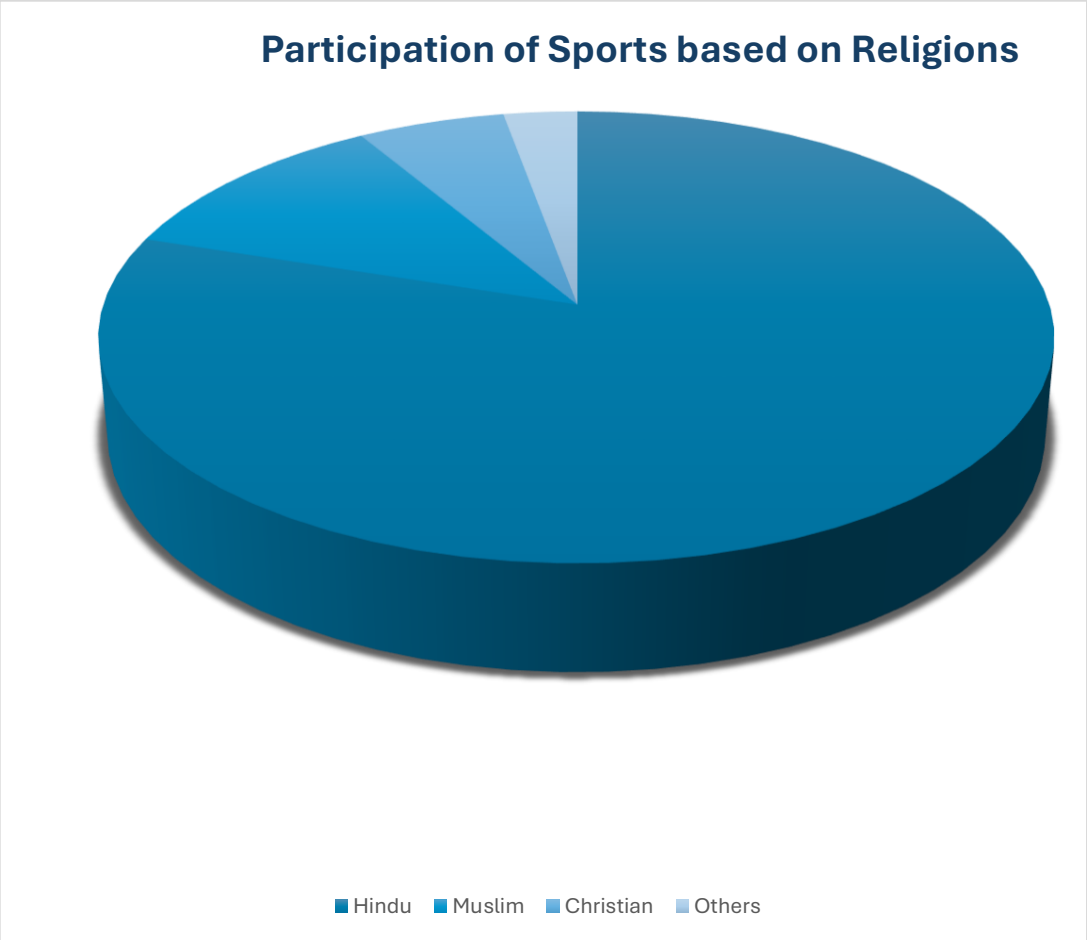


INTERPRETATIONS: -

- ✓ Hindus show higher engagement in physical activity.
- ✓ Muslims exhibit moderate physical activity levels.
- ✓ Christians & others have lower representation.

5.10 Participation in Sports/Exercises based on the Religion

Religion	Yes	No
Hindu	28	52
Muslim	4	10
Christian	2	1
Others	1	2
Total	35	65

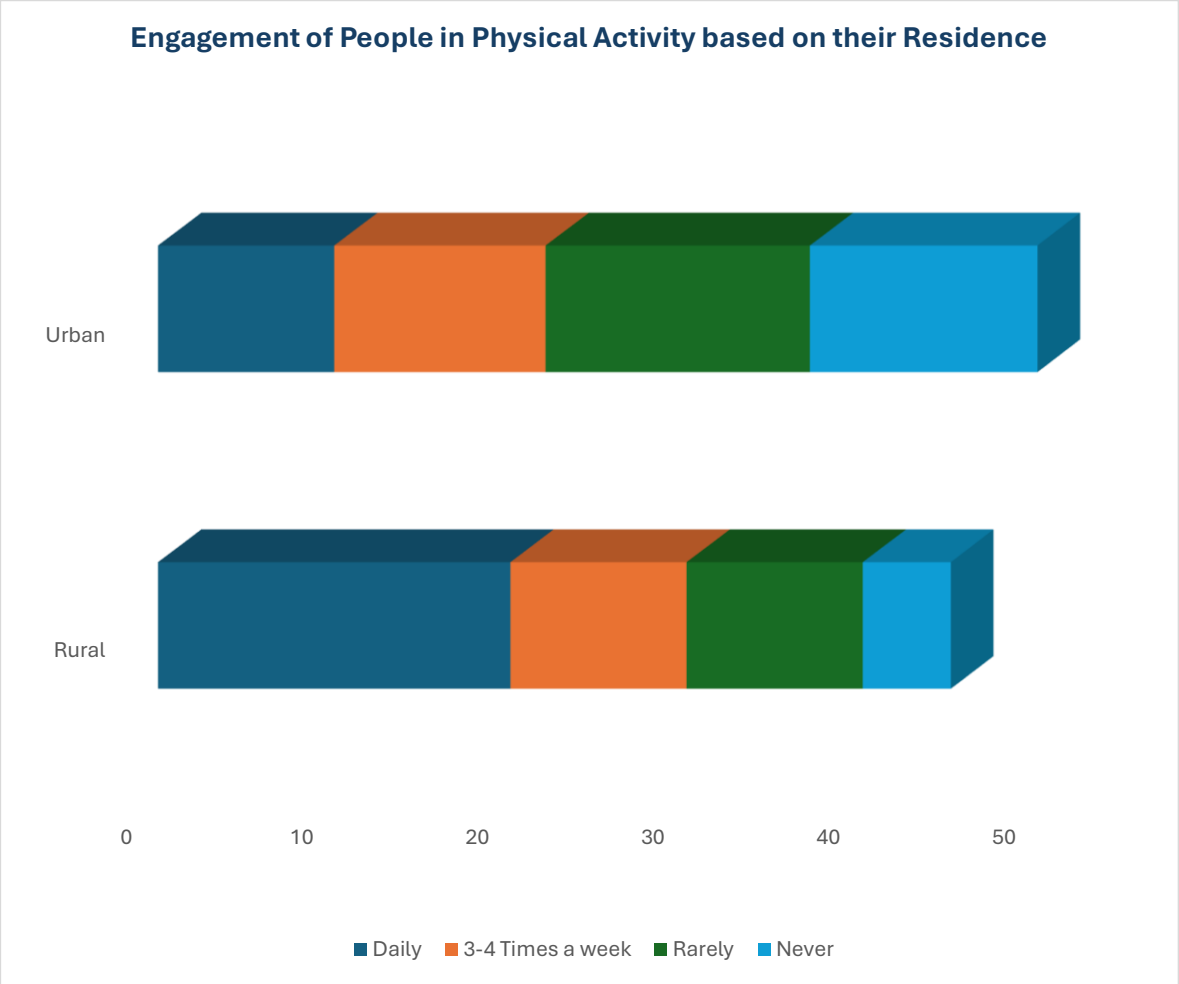


INTERPRETATIONS: -

- ✓ Hindu (80%): Reflecting the majority population in India, with a participation total (35%) in Sports
- ✓ Muslims (14%): Participation Rate is approx. 28.6%.
- ✓ Christian (3%): Higher participation rate at 66.7%; possibly due to cultural factors and community programs.

5.11 Engagement of people in Physical Activities according to their Residence

Residence	Daily	3-4 Times a week	Rarely	Never
Rural	20	10	10	5
Urban	10	12	15	13
Total	30	22	25	18

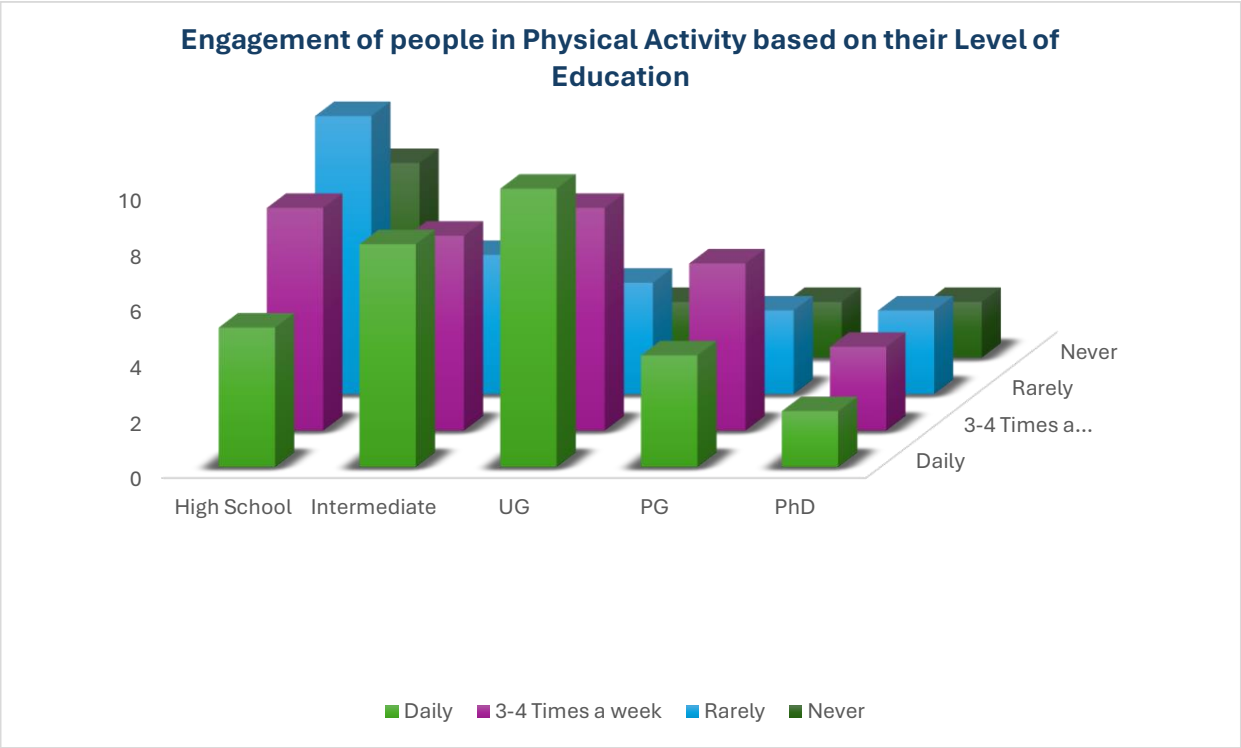


INTERPRETATIONS: -

- ✓ **Daily activity**: 40% rural vs. 20% urban
- ✓ **3–4 times/week**: 20% rural vs. 24% urban
- ✓ **Rarely active**: 20% rural vs. 30% urban
- ✓ **Never active**: 10% rural vs. 26% urban

5.12 Engagement of Physical Activity among people based on their level of Education

Level of Education	Daily	3-4 Times a week	Rarely	Never
High School	5	8	10	7
Intermediate	8	7	5	2
UG	10	8	4	2
PG	4	6	3	2
PhD	2	3	3	2
Total	29	32	25	15



INTERPRETATIONS: -

- ✓ High School graduates (30 participants): A significant portion engages in physical activity, with 10 rarely, and 7 never participating. Only 5 engage daily.
- ✓ Intermediate level (20 participants): There is a balanced distribution, with 6 engaging daily, 7 participating 3-4 times a week.
- ✓ Undergraduate: This shows higher engagement.
- ✓ Postgraduate: Moderate engagement is observed.
- ✓ PhD holders: Lower engagement levels are noted.

CHI- SQUARE TESTS

To check the Association between Engagement of people in Physical Activity with Gender: -

1. DATA:

The data is taken from the table "Gender wise distribution across physical activity":

Gender	Daily	3–4 Times a Week	Rarely	Never	Total
Male	18	7	18	5	48
Female	12	15	7	3	37
Total	**30**	**22**	**25**	**8**	**85**

2. HYPOTHESIS:

Null Hypothesis (H0): There is no association between gender and the frequency of engagement in physical activity.

Alternative Hypothesis (H1): There is an association between gender and the frequency of engagement in physical activity.

3. CALCULATE EXPECTED FREQUENCIES:

To calculate the expected frequencies, we use the formula:

Expected Frequency = (Row Total * Column Total) / Grand

Total Here are the expected frequencies for each cell:

* Expected Frequency (Male, Daily) = (48 * 30) / 85 = 16.94

* Expected Frequency (Male, 3-4 Times a week) = (48 * 22) / 85 = 12.49

* Expected Frequency (Male, Rarely) = (48 * 25) / 85 = 14.12

* Expected Frequency (Male, Never) = (48 * 8) / 85 = 4.52

* Expected Frequency (Female, Daily) = (37 * 30) / 85 = 13.06

* Expected Frequency (Female, 3-4 Times a week) = (37 * 22) / 85 = 9.58

* Expected Frequency (Female, Rarely) = (37 * 25) / 85 = 10.88

* Expected Frequency (Female, Never) = (37 * 8) / 85 = 3.48

4. CALCULATE THE CHI-SQUARE TEST STATISTIC:

The Chi-Square test statistic is calculated using the formula:

$X^2 = \Sigma$ ((Observed Frequency - Expected Frequency) ^{2} / Expected Frequency)

$X^2 = ((18 - 16.94)^2 / 16.94) + ((7 - 12.49)^2 / 12.49) + ((18 - 14.12)^2 / 14.12) + ((5 - 4.52)^2 / 4.52)$
$+$
$((12 - 13.06)^2 / 13.06) + ((15 - 9.58)^2 / 9.58) + ((7 - 10.88)^2 / 10.88) + ((3 - 3.48)^2 / 3.48)$

$X^2 = (0.11) + (2.40) + (1.06) + (0.05) + (0.09) + (3.02) + (1.36) + (0.07)$

$X^2 = 8.16$

5. DETERMINE THE DEGREES OF FREEDOM:

The Degrees of Freedom (DF) are calculated as:

DF= (Number of Rows - 1) * (Number of Columns - 1)

DF= (2 - 1) * (4 - 1) = 1 * 3 = 3

6. FIND THE P-VALUE:

Using a Chi-Square distribution table or a statistical calculator, with $X^2 = 8.16$ and df = 3, the pvalue is approximately 0.043.

7. MAKE A DECISION:

We compare the p-value to the significance level (α). A common significance level is 0.05.

Since 0.043 < 0.05, **WE REJECT THE NULL HYPOTHESIS.**

8. INTERPRETATION:

- ✓ The Chi-Square test results indicate that there is a statistically significant association between gender and the frequency of engagement in physical activity. In simpler terms, men and women show different patterns in how often they engage in physical activity.
- ✓ It's important to note that while the test shows an association, it doesn't explain why this difference exists. Further research could explore the factors contributing to these gender-based differences in physical activity levels, such as cultural norms, access to resources, or personal preferences.

To check the Association of Achieving Fitness goals to people according to their age Groups: -

1. DATA:

The following table shows the distribution of fitness goals among different age groups (assumed data based on typical survey patterns):

Age Group	Weight Loss	Muscle Gain	Flexibility	Others	Total
13-18	8	4	6	2	20
19-25	15	10	4	1	30
26-45	14	8	2	1	25
46-65	10	2	3	0	15
Total	**47**	**24**	**15**	**4**	**90**

2. HYPOTHESIS:

Null Hypothesis (H_0): There is no association between age group and fitness goals.

Alternative Hypothesis (H_1): There is an association between age group and fitness goals.

3. EXPECTED FREQUENCIES:

Age Group	Weight Loss	Muscle Gain	Flexibility	Others
13-18	10.44	5.33	3.33	0.89
19-25	15.67	8.00	5.00	1.33
26-45	13.06	6.67	4.17	1.11
46-65	7.83	4.00	2.50	0.67

4. CHI-SQUARE TEST STATISTIC:

The test statistic is calculated as:

$$\chi 2=\sum((Oi-Ei)^2/ Ei)$$

After performing the calculations, the chi-square value is:

$$\chi 2=9.97$$

5. DEGREES OF FREEDOM: DF = (4−1) (4−1) =3×3=9

6. P- VALUE:

With $\chi 2$ =9.97 and DF =9, the **p-value ≈ 0.35**

7. CONCLUSION:

Since the p-value is greater than 0.05, we **FAIL TO REJECT THE NULL HYPOTHESIS**.

8. INTERPRETATION:

- ✓ The chi-square test results indicate that there is no statistically significant association between age group and fitness goals. This means that individuals from different age groups in the sample share similar fitness aspirations, such as weight loss, muscle gain, or improved flexibility. Age does not appear to be a strong determining factor in shaping fitness goals.

To check the Association between Participations of people in Physical Activity according to their age Groups: -

1. DATA PREPARATION:

Age Group	Yes	No	Total
13–18	15	3	18
19–25	25	5	30
26–45	18	10	28
46–65	6	27	33
Total	64	45	109

2. HYPOTHESIS:

Null Hypothesis (H0): There is no association between age group and participation in sports.
Alternative Hypothesis (H1): There is an association between age group and participation in sports.

3. CALCULATE EXPECTED FREQUENCIES:

To calculate the expected frequencies, we use the formula:
Expected Frequency = (Row Total * Column Total) / Grand Total Here are the expected frequencies for each cell:
* Expected Frequency (13-18, Yes) = (18 * 64) / 109 = 10.59 * Expected Frequency (13-18, No) = (18 * 45) / 109 = 7.41
* Expected Frequency (19-25, Yes) = (30 * 64) / 109 = 17.61
* Expected Frequency (19-25, No) = (30 * 45) / 109 = 12.39
* Expected Frequency (26-45, Yes) = (28 * 64) / 109 = 16.42
* Expected Frequency (26-45, No) = (28 * 45) / 109 = 11.58
* Expected Frequency (46-65, Yes) = (33 * 64) / 109 = 19.36
* Expected Frequency (46-65, No) = (33 * 45) / 109 = 13.64

4. CALCULATE THE CHI-SQUARE TEST STATISTIC:

The Chi-Square test statistic (X^2) is calculated using the formula:
$X^2 = \Sigma$ ((Observed Frequency - Expected Frequency) ^{2} / Expected Frequency)
$X^2 = 35.05$

5. DETERMINE THE DEGREES OF FREEDOM:

DF= (4 - 1) * (2 - 1) = 3 * 1 = 3

6. FIND THE P-VALUE:

Using a Chi-Square distribution table or a statistical calculator, with $X^2 = 35.05$ and DF = 3, the p-value is approximately 0.000001 (or very close to 0).

7. MAKE A DECISION:

We compare the p-value to the significance level (α). A common significance level is 0.05. Since 0.000001 < 0.05, **WE REJECT THE NULL HYPOTHESIS**.

8. INTERPRETATION:

- ✓ The Chi-Square test results indicate that there is a statistically significant association between age group and participation in sports. In simpler terms, different age groups show different patterns of sports participation. The very small p-value suggests strong evidence against the idea that there's no relationship between age and sports participation in your sample.

To check the Association of Utilization of any Fitness Tracking Device or app according to Gender: -

1. DATA:

The data used for this test is taken from the table "Gender distribution of utilization of any fitness tracking device or apps among people."

Gender	Smartwatch	Fitness Tracker	Mobile Apps	Others	Total
Male	18	8	10	4	40
Female	22	12	14	12	60
Total	**40**	**20**	**24**	**16**	**100**

2. HYPOTHESIS:

Null Hypothesis (H_0): There is no association between gender and the type of fitness tracking device or app used.

Alternative Hypothesis (H_1): There is an association between gender and the type of fitness tracking device or app used.

3. EXPECTED FREQUENCIES:

Gender	Smartwatch	Fitness Tracker	Mobile Apps	Others
Male	(40×40)/100 = 16.00.	(40×20)/100 = 8.00	(40×24)/100 = 9.60	(40×16)/100 = 6.40
Female	(60×40)/100 = 24.00	(60×20)/100 = 12.00	(60×24)/100 = 14.40	(60×16)/100 = 9.60

4. CHI-SQUARE TEST STATISTIC:

$\chi 2 = 1.9447$

5. DEGREE OF FREEDOM:

DF= (2−1)(4−1) =3

6. P-VALUE:

With $\chi 2$=1.9447 and DF =3, the p-value is approximately **0.584**.

7. CONCLUSION:

Since the p-value (0.584) is greater than the significance level (0.05), we **FAIL TO REJECT THE NULL HYPOTHESIS**.

8. INTERPRETATIONS:

- ✓ The Chi-Square test indicates that there is **NO STATISTICALLY SIGNIFICANT ASSOCIATION** between gender and the type of fitness tracking device or app used. In simpler terms, both males and females in the sample exhibit similar usage patterns across smartwatches, fitness trackers, mobile apps, and other tools. This suggests that gender does not significantly influence the preference for digital fitness monitoring methods among the surveyed participants.

CHAPTER 6: RESULTS AND DISCUSSION

The Analysis of Survey data from 100 individuals reveals several interesting patterns and associations related to sports and fitness behaviours. The results highlight how demographic factors such as age, gender, residence, education, and religion influence physical activity levels and preferences.

1. AGE-BASED OBSERVATIONS:

- The 19–25 age group showed both the highest daily engagement and the highest percentage of individuals who never exercise, indicating a polarization in fitness habits among young adults.
- Running emerged as the most popular physical activity across all age groups, especially among the 13–25 age segments.
- Participation in sports was highest in the 19–25 age group and significantly lower in the 46–65 age groups, suggesting a drop in engagement with increasing age.

2. GENDER-BASED TRENDS:

- Males were slightly more active overall, especially in daily physical activity and participation in sports.
- Females showed a stronger preference for yoga and a higher usage of fitness tracking devices such as smartwatches and mobile apps.
- The chi-square test showed a significant association between gender and frequency of physical activity ($p < 0.05$), indicating that fitness habits vary notably between males and females.

3. RESIDENCE AND EDUCATION:

- Rural participants were more likely to engage in daily physical activity than their urban counterparts, possibly due to lifestyle differences.
- Educational level also had an influence, with undergraduate and intermediate-level participants showing higher daily engagement.

4. RELIGION AND PARTICIPATION:

- Hindus formed the majority of the sample and showed higher participation in both daily activity and sports.
- Christians, though fewer in number, had a relatively higher rate of participation, possibly influenced by cultural or community factors.

5. FITNESS GOALS AND USAGE OF TECHNOLOGY:

- Weight loss and muscle gain were equally prioritized across genders and age groups.
- Fitness goals were not significantly associated with age ($p > 0.05$), suggesting that people of all ages have similar aspirations in fitness.
- Smart-watches were the most used tracking device, especially among females, though overall, no significant gender difference was found in usage patterns ($p > 0.05$).

6. STATISTICAL TESTING SUMMARY:

The chi-square tests revealed significant associations between:

1. Gender and frequency of physical activity.
2. Age group and sports participation.

No significant associations were found between:

1. Age and fitness goals.
2. Gender and type of fitness tracking device.

CHAPTER 7: CONCLUSION

This Minor Research Project, titled "A Statistical Analysis of Sports and Fitness Patterns Based on Survey Data," aimed to explore the relationships between demographic factors and physical activity behaviours among 100 participants.

THE ANALYSIS CONCLUDES THAT:

- Age and gender are significant determinants of physical activity frequency and type.
- Running and yoga are the most popular activities, while gymnastic participation remains minimal.
- Rural populations are more active on a daily basis than urban counterparts.
- There is no strong link between age and fitness goals, indicating that people of all ages are equally motivated by goals such as weight loss and muscle gain.
- Technology use for fitness tracking is common, especially smartwatches and mobile apps, though gender does not significantly influence the type of device used.
- The project successfully applied statistical tools such as the chi-square test to validate associations and used graphical methods for clearer interpretation. These methods helped turn raw survey data into actionable insights.

RECOMMENDATIONS:

- More inclusive fitness programs should be designed to engage less active groups, particularly females and older adults.
- Awareness campaigns can encourage urban dwellers to adopt more active lifestyles.
- Policymakers should leverage the popularity of mobile apps and wearables to promote fitness in digital spaces.

In Summary, this study underlines the value of statistical research in understanding fitness patterns and developing data-driven strategies for improving public health outcomes. The insights gained here can serve as a foundation for future, larger-scale studies and community interventions focused on physical wellness.

CHAPTER 8: EXPERIENCE AND DIFFICULTIES

This Minor Research Project, "A Statistical Analysis of Sports and Fitness Patterns Based on Survey Data," has proven to be an immersive and transformative experience, providing a comprehensive introduction to the multifaceted world of statistical research. The journey, from the initial conceptualization of the research question to the final interpretation of the analysed data, was marked by a series of learning milestones and practical challenges that have significantly enriched my understanding of the research process. This project has not only deepened my appreciation for the power of statistical methodologies in uncovering meaningful insights but also instilled in me a greater awareness of the complexities and nuances inherent in conducting social science research.

The meticulous design of the survey questionnaire represented a critical initial phase that demanded a blend of creativity, analytical thinking, and a thorough understanding of the research domain. This process involved an extensive review of existing literature on sports, fitness, and related behavioural studies, which served as a foundation for identifying key variables and formulating relevant research questions. The translation of these research questions into clear, concise, and unbiased survey questions required careful consideration of question types (e.g., multiple-choice, Likert scale, open-ended), question wording, and question sequencing. A pilot study, conducted with a small group of individuals prior to the main data collection, proved to be an invaluable tool for refining the questionnaire, identifying potential ambiguities, and ensuring its overall clarity and effectiveness. This iterative process of questionnaire development underscored the importance of careful planning, attention to detail, and the value of pilot testing in minimizing measurement error and maximizing the validity of research findings.

The subsequent stage of data collection, while limited to a convenience sample of 100 participants, offered a practical introduction to the logistical and ethical considerations involved in engaging human subjects in research. The decision to employ convenience sampling, while primarily driven by constraints in time and resources, highlighted the inherent trade-offs between practicality and the representativeness of the sample. While this method facilitated efficient data collection, it also prompted critical reflections on the potential limitations in generalizing the findings to a broader population. Furthermore, the reliance on self-reported data underscored the importance of being cognizant of potential sources of bias, such as social desirability bias (the tendency of respondents to provide answers that they believe are socially acceptable) and recall bias (inaccuracies in responses due to memory limitations). To mitigate these biases, various strategies were implemented, including ensuring the anonymity of participants, emphasizing the confidentiality of responses, and stressing the importance of honest and accurate reporting. The data collection process also provided valuable experience in communicating with participants, addressing their concerns, and ensuring their informed consent.

Throughout the project, several challenges were encountered that provided valuable learning experiences. The relatively small sample size of 100 participants-imposed limitations on the statistical power of the analyses, potentially hindering the ability to detect subtle but meaningful

effects. The use of convenience sampling, as previously discussed, raised concerns about the generalizability of the findings and the potential for selection bias. The reliance on self-reported data introduced the possibility of various biases that could have affected the accuracy of the results. Time constraints presented challenges in conducting a more comprehensive analysis, exploring additional research questions, and delving deeper into the nuances of the data. Furthermore, the initial lack of familiarity with certain statistical software functions required a significant investment of time in learning and mastering these tools.

Despite these challenges, this research project has been an immensely valuable and enriching experience. It has provided practical insights into the entire research process, from the initial formulation of research questions and the design of data collection instruments to the analysis and interpretation of data. The limitations encountered throughout the project have served as important reminders of the complexities inherent in research and the need for careful consideration of methodological choices. This experience has not only reinforced the importance of statistical rigor, critical thinking, and effective communication in research but has also ignited a deeper passion for pursuing further investigations in the field of health and fitness, with a focus on addressing some of the limitations encountered in this project and contributing to a more robust and nuanced understanding of this important area.

CHAPTER 9: QUESTIONNAIRE

DEMOGRAPHIC: -

What is your Age?

a. 13-18 [ ] b. 18-25 [ ]c. 25-45 [ ] d. 45-65 [ ] e. 65+ [ ]

What is your Gender?

a. Male [ ] b. Female [ ] c. Non-binary [ ] d. Others [ ]

What is your Religion?

a. Hindu [ ] b. Muslim [ ] c. Christian [ ] d. Others [ ]

What is your Residence?

a. Rural [ ] b. Urban [ ]

What is your level of Education?

a. High school [ ] b. Inter [ ] c. UG [ ] d. PG [ ] e. PhD [ ]

What is your Family Income?

a. Below 30K [ ] b. 30K-50K [ ] c. 50K-75K [ ] d. More than 75K [ ]

PHYSICAL ACTIVITY: -

How Often do you Engage in Physical Activity?

a. Daily [ ] b. 3-4 times a week [ ] c. Rarely [ ] d. Never [ ]

What Type of Physical Activity do you enjoy the most?

a. Running [ ] b. Swimming [ ] c. Yoga [ ] d. Gymnastics [ ] e. Others[]

Which Physical Activity do you do Every Day?

a. Cycling [ ] b. Running [ ] c. Yoga [ ] d. Gymnastics [ ] e. Others [ ]

FITNESS GOALS: -

What are your Fitness Goals?

a. Weight loss [ ] b. Muscle gain [ ] c. Improved flexibility [ ] d. Others [ ]

How Important is Achieving your Fitness Goals to you?

a. Very important [ ] b. Somewhat important [ ] c. Not very imp [ ] d. Not at all [ ]

What Type of Impact do you receive from your Physical Activity?

a. Low impact [ ] b. Very high impact [ ] c. I'm not sure [ ] d. Others [ ]

SPORTS PARTICIPATION: -

Do you Participate in any Sports or Exercises Competitively?

a. Yes [ ] b. No [ ]

If Yes, Which Sports or Physical Activity do you participate in Competitively?

a. Team sports [ ] b. Individual sports [ ] c. Strength sports [ ] d. Others [ ]

Do your Institute Celebrates Sports Day every year?

a. Yes [ ] b. No [ ]

INJURY AND RECOVERY: -

Have you ever Suffered any Injury while participating in Sports or Physical Activity? a. Yes [ ] b. No [ ]

If Yes, what type of Injury do you Suffer?

a. Muscle strain [ ] b. Joint injury [ ] c. Bone fracture [ ] d. Others [ ]

How do you Recover from Injury?

a. Rest and Recovery [ ] b. Physical therapy [ ] c. Surgery [ ] d. Others [ ]

Do you use any Fitness Tracking device or apps?

a. Smartwatch [ ] b. Fitness tracker [ ] c. Mobile apps [ ] d. Others [ ]

Have you ever undergone Physical Therapy and Rehabilitation?

a. Yes [ ] b. No [ ]

If Yes, What Types of Recovery Techniques were used?

a. Stretching [ ] b. Strengthening [ ] c. Mobilization [ ] d. Others [ ]